MSG No. 127.391 $P4'/mbm'$ [Type III, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `4'/m..`

No.	position	mapping
1	$[0, 0, 0]$	`[1,4,5,8,9,10,13,14]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,3,6,7,11,12,15,16]`

Table 2: Wyckoff site: `2b`, site symmetry: `4'/m..`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,4,5,8,9,10,13,14]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,3,6,7,11,12,15,16]`

Table 3: Wyckoff site: `2c`, site symmetry: `m.m'm'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,4,5,8,11,12,15,16]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,3,6,7,9,10,13,14]`

Table 4: Wyckoff site: `2d`, site symmetry: `m.m'm'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,4,5,8,11,12,15,16]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[2,3,6,7,9,10,13,14]`

Table 5: Wyckoff site: `4e`, site symmetry: `4'..`

No.	position	mapping
1	$[0, 0, z]$	`[1,4,9,10]`
2	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[2,3,11,12]`
3	$[0, 0, -z]$	`[5,8,13,14]`
4	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[6,7,15,16]`

Table 6: Wyckoff site: `4f`, site symmetry: `2.m'm'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, z\right]$	`[1,4,15,16]`
2	$\left[\frac{1}{2}, 0, -z\right]$	`[2,3,13,14]`
3	$\left[0, \frac{1}{2}, -z\right]$	`[5,8,11,12]`
4	$\left[\frac{1}{2}, 0, z\right]$	`[6,7,9,10]`

Table 7: Wyckoff site: `4g`, site symmetry: `m.2'm'`

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, 0\right]$	`[1,8,11,16]`
2	$\left[x+\frac{1}{2},\, -x,\, 0\right]$	`[2,7,10,13]`
3	$\left[\frac{1}{2}-x,\, x,\, 0\right]$	`[3,6,9,14]`
4	$\left[-x,\, \frac{1}{2}-x,\, 0\right]$	`[4,5,12,15]`

Table 8: Wyckoff site: `4h`, site symmetry: `m.2'm'`

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, \frac{1}{2}\right]$	`[1,8,11,16]`
2	$\left[x+\frac{1}{2},\, -x,\, \frac{1}{2}\right]$	`[2,7,10,13]`
3	$\left[\frac{1}{2}-x,\, x,\, \frac{1}{2}\right]$	`[3,6,9,14]`
4	$\left[-x,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	`[4,5,12,15]`

Table 9: Wyckoff site: `8i`, site symmetry: `m..`

No.	position	mapping
1	$[x,\, y,\, 0]$	`[1,8]`
2	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, 0\right]$	`[2,7]`
3	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, 0\right]$	`[3,6]`
4	$[-x,\, -y,\, 0]$	`[4,5]`
5	$[-y,\, x,\, 0]$	`[9,14]`
6	$[y,\, -x,\, 0]$	`[10,13]`
7	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, 0\right]$	`[11,16]`
8	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, 0\right]$	`[12,15]`

Table 10: Wyckoff site: `8j`, site symmetry: `m..`

No.	position	mapping
1	$\left[x,\, y,\, \frac{1}{2}\right]$	`[1,8]`
2	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	`[2,7]`
3	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}\right]$	`[3,6]`
4	$\left[-x,\, -y,\, \frac{1}{2}\right]$	`[4,5]`
5	$\left[-y,\, x,\, \frac{1}{2}\right]$	`[9,14]`
6	$\left[y,\, -x,\, \frac{1}{2}\right]$	`[10,13]`
7	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}\right]$	`[11,16]`
8	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	`[12,15]`

Table 11: Wyckoff site: `8k`, site symmetry: `..m'`

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, z\right]$	`[1,16]`
2	$\left[x+\frac{1}{2},\, -x,\, -z\right]$	`[2,13]`

continued ...

Table 11

No.	position	mapping
3	$\left[\frac{1}{2}-x,\ x,\ -z\right]$	[3,14]
4	$\left[-x,\ \frac{1}{2}-x,\ z\right]$	[4,15]
5	$\left[-x,\ \frac{1}{2}-x,\ -z\right]$	[5,12]
6	$\left[\frac{1}{2}-x,\ x,\ z\right]$	[6,9]
7	$\left[x+\frac{1}{2},\ -x,\ z\right]$	[7,10]
8	$\left[x,\ x+\frac{1}{2},\ -z\right]$	[8,11]

Table 12: Wyckoff site: 16l, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[2]
3	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ -z\right]$	[3]
4	$[-x,\ -y,\ z]$	[4]
5	$[-x,\ -y,\ -z]$	[5]
6	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z\right]$	[6]
7	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$	[7]
8	$[x,\ y,\ -z]$	[8]
9	$[-y,\ x,\ z]$	[9]
10	$[y,\ -x,\ z]$	[10]
11	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ -z\right]$	[11]
12	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ -z\right]$	[12]
13	$[y,\ -x,\ -z]$	[13]
14	$[-y,\ x,\ -z]$	[14]
15	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z\right]$	[15]
16	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z\right]$	[16]